

Data Analysis, Visualization, and Communication

Laboratórios de Sistemas e Serviços

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Exercises

Exercise 1: The DIKW Pyramid

Explain, using a practical example from an IoT weather station, the difference between: 1. **Data** 2. **Information** 3. **Knowledge** 4. **Wisdom**

Exercise 2: Data Loading and Manipulation

Using the dataset/`iris.csv` file (or any other dataset available in the repository):

1. Load the dataset using **Pandas**.
 2. Display the first 5 rows and the column names.
 3. Calculate the mean and standard deviation of one of the numerical columns.
 4. Filter the data to show only rows where a specific variable is above a certain threshold.
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Exercise 3: Handling Missing Data

You have the following sensor readings with missing values: [22.1, 23.5, NaN, 22.8, 23.0, NaN, 24.2]

1. Calculate the **Mean** of the available data.
 2. Replace the NaN values with the Mean (Imputation).
 3. Why might using the **Median** be better if the data had extreme outliers?
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Exercise 4: Data Visualization

Using **Seaborn** or **Matplotlib**:

1. Create a **Histogram** of a numerical variable to see its distribution.
 2. Create a **Scatter Plot** between two numerical variables to see if they are correlated.
 3. Create a **Box Plot** to compare a numerical variable across different categories.
 4. Save your plots as `.pdf` (Vector) and `.png` (Raster). Compare the quality when zooming in.
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Exercise 5: Correlation

1. What does a correlation coefficient of **-0.85** tell you about the relationship between two variables?
 2. Is it possible for two variables to have a strong relationship but zero Pearson correlation?
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Exercise 6: Data Communication and Sharing

1. What is the difference between **Serialization** and **Deserialization**?
2. Why is a **Schema** important when sharing data between different systems?
3. Match the protocol to its description:
 - **REST**
 - **MQTT**
 - **WebSockets**

A. Publish/Subscribe model for IoT. B. Standard request/response using HTTP. C. Real-time, full-duplex communication.
4. Why should you use **ISO 8601** for timestamps?